

Coupled 0-D/1-D Fontan Model Technical Reference

Standardized model definition, equations, segments, and free parameters

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1 Coupled 0-D/1-D Fontan Model Technical Reference

This document is generated from repository sources by `scripts/docs/build_model_reference_pdfs.py`. Edit the model config, implementation notes, schematic, or this generator, then regenerate the markdown and PDF together.

1.1 Model Construction

1.1.1 Scope and Status

Executable Task 012 prototype with true 1-D aorta and TCPC segments inserted into the closed loop; not accepted as a periodic calibrated model yet.

Task 010 provides a local fixed three-cell true 1-D vessel prototype with area and face-flow states, nonlinear momentum, a pressure-area wall law, pressure/flow coupling, and tested Jacobian assembly.

Task 011 provides open-loop reference specifications for aorta, TCPC, and combined aorta-TCPC 1-D submodels using tracked Aramburu/Nektar geometry, measured inflows, reference outputs, clinical comparison targets, and documented validation gates.

1.1.5 Scenario Configs

- models/coupled_0d_1d/configs/fontan_coupled_0d_1d_baseline.jsonc
- models/coupled_0d_1d/configs/fontan_coupled_0d_1d_fenestration.jsonc
- models/coupled_0d_1d/configs/fontan_coupled_0d_1d_lpa_obstruction.jsonc
- models/coupled_0d_1d/configs/fontan_coupled_0d_1d_smoke.jsonc
- models/coupled_0d_1d/configs/fontan_coupled_0d_1d_vasodilation.jsonc

1.1.6 Pressure Nodes

- atrial
- cavity
- aao
- bca
- lcca
- lsa
- upper_art
- upper_ven
- dao
- lower_art
- lower_ven
- svc
- ivc
- rpa
- lpa
- coupled_aao_arch
- coupled_dao_arch
- coupled_bca_arch
- coupled_lcca_arch
- coupled_lsa_arch
- coupled_dao_out
- coupled_bca_out
- coupled_lcca_out
- coupled_svc_tpc
- coupled_ivc_tpc
- coupled_rpa_tpc
- coupled_lpa_tpc

1.1.7 Block Type Counts

Block type	Count
aortic_arch_total_pressure_junction_block	1
c_block	14
fixed_3cell_1d_log_area_vessel_block	7
fixed_6cell_tapered_1d_log_area_vessel_block	1
rc_block	13
rcr_block	2
spherical_cavity_block	1
tpc_characteristic_total_pressure_junction_block	1
time_varying_elastance_atrium_block	1
valve_rl_block	2

1.2 Governing Equations

The sign convention follows PhysioBlocks local block fluxes. For a two-node flow element, local node 1 is the first node listed in the config and local node 2 is the second node listed in the config.

1.2.1 Nodal Conservation

At every pressure node, the algebraic/differential network residual is the sum of all block flux contributions attached to that node:

$$\sum_{b \in \mathcal{B}(i)} Q_{b,i} = 0.$$

Storage blocks contribute pressure derivatives to this same residual, so closed-loop volume conservation is enforced through the connected block equations rather than through prescribed boundary flow.

1.2.2 Passive Compliance Block

For a `c_block` at pressure node `P` with capacitance `C`:

$$Q = -C \frac{dP}{dt}.$$

The saved stored volume is proportional to pressure in the local linear compliance approximation:

$$V = CP.$$

1.2.3 Pure RC Resistor Convention

This repository uses `rc_block` with zero capacitance as a pure resistive link. `PhysioBlocks` defines the local fluxes as:

$$Q_1 = \frac{P_2 - P_1}{R},$$
$$Q_2 = \frac{P_1 - P_2}{R} - C \frac{dP_2}{dt}.$$

When `C` = 0, the block is used as a pure resistor. To represent an upstream-to-downstream path with positive physical flow from upstream to downstream, the configs assign local node 2 to the upstream pressure and local node 1 to the downstream pressure.

1.2.4 Hydraulic R-L Link

The local quasi-vessel R-L element uses positive internal flow from local node 1 to local node 2:

$$L \frac{dQ}{dt} + RQ = P_1 - P_2,$$

$$Q_1 = -Q, \quad Q_2 = Q.$$

This is the repeated segment equation used by the quasi 0-D/1-D chains.

1.2.5 Valve R-L Block

The R-L valve block has local positive flow from node 1 to node 2 and switches conductance according to flow direction:

$$L \frac{dQ}{dt} + P_2 - P_1 + R(Q)Q = 0,$$

$$R(Q) = \begin{cases} 1/G_f, & Q > 0, \\ 1/G_b, & Q < 0, \end{cases}$$

$$Q_1 = -Q, \quad Q_2 = Q.$$

G_f is the forward conductance and G_b is the backward conductance.

1.2.6 Pulmonary RCR Windkessel

For an `rcr_block` with inlet pressure P_1 , outlet pressure P_2 , middle pressure P_m , proximal resistance R_1 , distal resistance R_2 , and compliance C :

$$Q_1 = \frac{P_m - P_1}{R_1},$$

$$Q_2 = \frac{P_m - P_2}{R_2},$$

$$\frac{P_1 - P_m}{R_1} + \frac{P_2 - P_m}{R_2} - C \frac{dP_m}{dt} = 0.$$

1.2.7 Active Atrium

The active atrium is a one-node time-varying elastance chamber:

$$E(t) = E_{min} + (E_{max} - E_{min})a(t),$$

$$V_a(t) = V_{0,a} + \frac{P_a(t) - P_{ext}}{E(t)},$$

$$Q_a = -\frac{dV_a}{dt}.$$

The activation $a(t)$ is a raised-cosine pulse over the configured start, peak, and end phase of the cardiac cycle.

1.2.8 Spherical Ventricular Cavity

The active ventricular cavity stores volume through the spherical cavity displacement y , reference radius R_0 , and wall thickness d_0 :

$$V(y) = \frac{4\pi}{3} \left[R_0 \left(1 + \frac{y}{R_0} \right) - \frac{d_0}{2 \left(1 + \frac{y}{R_0} \right)^2} \right]^3,$$

$$Q_v = -\frac{dV(y)}{dt}.$$

The pressure-displacement relation comes from the configured PhysioBlocks spherical dynamics, velocity law, passive rheology, and active macro-Huxley submodels. The corresponding free parameters are listed in the parameter inventory below with the `cavity.*` prefix.

1.3 Local True 1-D Prototype Equations

Task 010 adds a local fixed three-cell true 1-D vessel prototype in `fontan_blocks.one_d`. Task 012 also adds a log-area executable variant for closed-loop coupling and a fixed six-cell tapered log-area variant for the composite LPA. These forms solve the same 1-D finite-volume equations.

For vessel length L and $N = 3$ finite-volume cells:

$$\Delta x = \frac{L}{3}.$$

The area states are cell-centered $A_i(t)$ for $i = 1, 2, 3$ and the flow states are staggered face flows $Q_j(t)$ for $j = 0, 1, 2, 3$. In the closed-loop executable block, the nonlinear state is $g_i(t) = \log A_i(t)$ and $A_i(t) = \exp(g_i(t))$.

The nonlinear wall law is:

$$P(A) = P_{\text{ext}} + \beta(\sqrt{A} - \sqrt{A_0}),$$

with wave speed:

$$c(A) = \sqrt{\frac{A}{\rho} \frac{dP}{dA}}, \quad \frac{dP}{dA} = \frac{\beta}{2\sqrt{A}}.$$

The finite-volume continuity residual is:

$$R_{A_i} = \frac{A_i^{n+1} - A_i^n}{\Delta t} + \frac{Q_i^{n+1/2} - Q_{i-1}^{n+1/2}}{\Delta x}.$$

The face momentum residual is:

$$R_{Q_j} = \frac{Q_j^{n+1} - Q_j^n}{\Delta t} + \left[\frac{\partial}{\partial x} \left(\alpha \frac{Q^2}{A} \right) \right]_j^{n+1/2} + \frac{A_{f,j}^{n+1/2}}{\rho} \left[\frac{\partial P}{\partial x} \right]_j^{n+1/2} + \kappa \frac{Q_j^{n+1/2}}{A_{f,j}^{n+1/2}}.$$

Boundary pressure gradients use half-cell distances and PhysioBlocks terminal fluxes use $Q_{\text{node}1} = -Q_0$ and $Q_{\text{node}2} = Q_3$.

The log-area Jacobian columns use the chain rule:

$$\frac{\partial R}{\partial g_i} = \frac{\partial R}{\partial A_i} A_i.$$

Prototype free parameters are `length` in m, `reference_area` in m^2 , `wall_stiffness` in Pa m^{-1} , `external_pressure` in Pa, `density` in kg m^{-3} , `friction_coefficient` in $\text{m}^2 \text{s}^{-1}$, and dimensionless `momentum_correction`.

The complete Task 010 equation notes, validation status, saved quantities, and limitations are maintained in `docs/one_d_numerics.md`.

1.4 Open-Loop 1-D Reference Specs

Task 011 adds strict-JSON reference specs for three open-loop 1-D submodels:

- `models/coupled_0d_1d/configs/submodel_aorta_1d_openloop.jsonc`
- `models/coupled_0d_1d/configs/submodel_tpc_1d_openloop.jsonc`
- `models/coupled_0d_1d/configs/submodel_aorta_tpc_1d_openloop.jsonc`

These files are generated by `scripts/modeling/derive_1d_geometry.py`. They bind the tracked patient-specific geometry, measured inflows, Nektar reference domain files, paper/comparison reference tables, clinical comparison targets, and validation tolerances. They are not PhysioBlocks launcher configs and they do not promote a coupled closed-loop model.

The aorta spec contains four source segments: ascending aorta, thoracic aorta, brachiocephalic, and left carotid. A normal LSA branch is not added because it is absent from the patient-specific geometry table.

The TCPC spec contains five source segments: IVC, RPA, LPA I, LPA II, and SVC. The combined spec maps aorta domains 1-4 and TCPC domains 5-9 in the tracked combined Nektar output.

Validation is run with:

```
python3 scripts/calibration/validate_1d_submodels.py
```

The current report is `models/coupled_0d_1d/reference_outputs/openloop_1d_validation.json` and all three open-loop specs pass the current reference screen. The detailed Task 011 policy, inputs, tolerances, and limitations are maintained in `docs/openloop_1d_submodels.md`.

1.5 Task 012 Closed-Loop Prototype

Task 012 generates executable closed-loop configs with:

```
python3 scripts/modeling/build_coupled_configs.py
python3 scripts/modeling/build_coupled_configs.py --check
```

The generator starts from the full 0-D scenario configs, removes the aortic and TCPC shortcut blocks, and inserts seven three-cell true 1-D log-area vessel blocks, one six-cell tapered LPA block, one massless aortic total-pressure junction, one massless wall-pressure-blended dissipative TCPC total-pressure junction, and three downstream aortic residual interface loss blocks. The calibrated full 0-D LSA terminal branch is retained as a non-1-D aortic outlet because no patient-specific LSA 1-D geometry is available.

Aortic 1-D blocks:

Block	Nodes	Source segment
<code>coupled_aao</code>	<code>aao -> coupled_aao_arch</code>	Ascending aorta
<code>coupled_dao</code>	<code>coupled_dao_arch -></code> <code>coupled_dao_out</code>	Thoracic aorta
<code>coupled_bca</code>	<code>coupled_bca_arch -></code> <code>coupled_bca_out</code>	Brachiocephalic
<code>coupled_lcca</code>	<code>coupled_lcca_arch -></code> <code>coupled_lcca_out</code>	Carotic left
<code>coupled_aortic_arch_junction</code>	AAo/DAo/BCA/LCCA/LSA ports	total-pressure junction

TCPC 1-D blocks and junction:

Block	Nodes	Source segment
<code>coupled_svc</code>	<code>svc -> coupled_svc_tcpc</code>	SVC
<code>coupled_ivc</code>	<code>ivc -> coupled_ivc_tcpc</code>	IVC
<code>coupled_rpa</code>	<code>coupled_rpa_tcpc -> rpa</code>	RPA
<code>coupled_lpa</code>	<code>coupled_lpa_tcpc -> lpa</code>	LPA I + LPA II
<code>coupled_tcpc_junction</code>	SVC/IVC/RPA/LPA branch ports	dissipative total-pressure junction

Residual aortic loss blocks preserve the full 0-D shortcut resistance not already represented by 1-D Poiseuille friction. For a retained path:

$$R_{\text{loss}} = \max(R_{\text{full0D}} - R_{\text{1D}}, 0).$$

The TCPC junction enforces algebraic mass balance with effective branch total-pressure compatibility:

$$Q_{\text{SVC}} + Q_{\text{IVC}} - Q_{\text{RPA}} - Q_{\text{LPA}} = 0,$$

$$H_k^* - H_{\text{RPA}}^* = 0, \quad k \in \{\text{SVC}, \text{IVC}, \text{LPA}\},$$

$$H_i^* = H_i + L_i,$$

$$H_i = wP_{\text{wall},i} + (1-w)P_{\text{node},i} + \frac{1}{2}\rho \left(\frac{Q_i}{A_i} \right)^2,$$

$$L_i = \frac{1}{2}\rho K \frac{q_{\text{out},i} \sqrt{q_{\text{out},i}^2 + \epsilon_Q^2}}{A_i^2}.$$

The current generated TCPC settings are `wall_pressure_weight = 0.75`, `loss_coefficient = 2.0`, and `characteristic_scale = 0.0`. The aortic junction remains a no-loss total-pressure coupler.

All generated coupled scenarios cap `time.step_size` at `0.00025 s` and `time.min_step` at `1.5625e-05 s`; the inherited full 0-D `0.002 s` step can make the first coupled nonlinear solve intractable.

The current startup smoke run completes `0.025 s` with no NaNs, no negative saved 1-D areas, near-zero aortic/TCPC junction mass residuals, passing TCPC balance, and bounded TCPC effective total-pressure spread around `0.36 mmHg`. The run is not accepted as physiological validation because atrium/ventricle balance and periodicity remain unproven. The tracked report is `models/coupled_0d_1d/reference_outputs/closed_loop_smoke_validation.json`.

A `2.0 s` baseline-derived diagnostic also completes with no NaNs, no negative 1-D areas, passing TCPC balance, and TCPC total-pressure spread around `0.36 mmHg`. It is not accepted as periodic validation because atrium/ventricle balance and cavity periodicity still fail.

1.6 Segment Inventory

Each row is a block or segment in the accepted baseline config. The parameter column lists explicit block fields and all config parameters sharing the block-name prefix.

Segment/block	Type	Local nodes	Free-parameter fields
cavity	spherical_cavity_block	1: cavity	disp = cavity.dynamics.disp; radius -> heart_radius; thickness -> heart_thickness; cav- ity.dynamics.damping_coef; cav- ity.dynamics.hyperelastic_cst; cavity.dynamics.vol_mass; cav- ity.rheology.active_law.activation; cav- ity.rheology.active_law.crossbridge_sti cav- ity.rheology.active_law.destruction_ra cav- ity.rheology.active_law.starling_abscis cav- ity.rheology.active_law.starling_ordin cav- ity.rheology.damping_parallel; cav- ity.rheology.series_stiffness; cav- ity.velocity_law.scheme_ts_hht backward_conductance -> valves.backward_conductance; valve_atrium.conductance; valve_atrium.inductance; valve_atrium.scheme_ts_flux backward_conductance -> valves.backward_conductance; valve_atrium.conductance; valve_atrium.inductance; valve_atrium.scheme_ts_flux capacitance_valve.capacitance pressure_external -> pleural.pressure; elastance_min -> ac- tive_atrium.elastance_min; elastance_max -> ac- tive_atrium.elastance_max; unstressed_volume -> ac- tive_atrium.unstressed_volume; activation_start -> ac- tive_atrium.activation_start; activation_peak -> ac- tive_atrium.activation_peak; activation_end -> ac- tive_atrium.activation_end; heartbeat_duration -> heartbeat_duration aao_compliance.capacitance bca_compliance.capacitance lcca_compliance.capacitance lsa_compliance.capacitance
valve_atrium	valve_rl_block	1: atrial; 2: cavity	
valve_arterial	valve_rl_block	1: cavity; 2: aao	
capacitance_valve	c_block	1: cavity	
active_atrium	time_varying_elastance_atrium_block	1: atrial; 2: cavity	
aao_compliance	c_block	1: aao	
bca_compliance	c_block	1: bca	
lcca_compliance	c_block	1: lcca	
lsa_compliance	c_block	1: lsa	

Segment/block	Type	Local nodes	Free-parameter fields
upper_ca1	c_block	1: upper_art	upper_ca1.capacitance
upper_cv1	c_block	1: upper_ven	upper_cv1.capacitance
dao_compliance	c_block	1: dao	dao_compliance.capacitance
lower_ca2	c_block	1: lower_art	lower_ca2.capacitance
lower_cv2	c_block	1: lower_ven	lower_cv2.capacitance
svc_compliance	c_block	1: svc	svc_compliance.capacitance
ivc_compliance	c_block	1: ivc	ivc_compliance.capacitance
rpa_compliance	c_block	1: rpa	rpa_compliance.capacitance
lpa_compliance	c_block	1: lpa	lpa_compliance.capacitance
upper_bca_to_ca1	rc_block	1: upper_art; 2: bca	capacitance -> zero_capacitance; upper_bca_to_ca1.resistance
upper_lcca_to_ca1	rc_block	1: upper_art; 2: lcca	capacitance -> zero_capacitance; upper_lcca_to_ca1.resistance
arch_lsa	rc_block	1: lsa; 2: coupled_lsa_arch	capacitance -> zero_capacitance; arch_lsa.resistance
upper_lsa_to_ca1	rc_block	1: upper_art; 2: lsa	capacitance -> zero_capacitance; upper_lsa_to_ca1.resistance
upper_rc1	rc_block	1: upper_ven; 2: upper_art	capacitance -> zero_capacitance; upper_rc1.resistance
upper_rv1	rc_block	1: svc; 2: upper_ven	capacitance -> zero_capacitance; upper_rv1.resistance
lower_ra4	rc_block	1: lower_art; 2: dao	capacitance -> zero_capacitance; lower_ra4.resistance
lower_rc2	rc_block	1: lower_ven; 2: lower_art	capacitance -> zero_capacitance; lower_rc2.resistance
lower_rv2	rc_block	1: ivc; 2: lower_ven	capacitance -> zero_capacitance; lower_rv2.resistance
right_lung	rcr_block	1: rpa; 2: atrial	pressure_mid = right_lung.pressure_mid; resistance_1 -> right_lung.resistance_1; resistance_2 -> right_lung.resistance_2; capacitance -> right_lung.capacitance
left_lung	rcr_block	1: lpa; 2: atrial	pressure_mid = left_lung.pressure_mid; resistance_1 -> left_lung.resistance_1; resistance_2 -> left_lung.resistance_2; capacitance -> left_lung.capacitance
fenestration	rc_block	1: atrial; 2: ivc	capacitance -> zero_capacitance; fenestration.resistance

Segment/block	Type	Local nodes	Free-parameter fields
coupled_aao	fixed_3cell_1d_log_area_vessel_block	coupled_aao_arch	length -> coupled_aao.length; reference_area -> coupled_aao.reference_area; wall_stiffness -> coupled_aao.wall_stiffness; external_pressure -> coupled_aao.external_pressure; density -> coupled_aao.density; friction_coefficient -> coupled_aao.friction_coefficient; momentum_correction -> cou-
coupled_dao	fixed_3cell_1d_log_area_vessel_block	coupled_dao_arch; 2: coupled_dao_out	length -> coupled_dao.length; reference_area -> coupled_dao.reference_area; wall_stiffness -> coupled_dao.wall_stiffness; external_pressure -> coupled_dao.external_pressure; density -> coupled_dao.density; friction_coefficient -> coupled_dao.friction_coefficient; momentum_correction -> cou-
coupled_bca	fixed_3cell_1d_log_area_vessel_block	coupled_bca_arch; 2: coupled_bca_out	length -> coupled_bca.length; reference_area -> coupled_bca.reference_area; wall_stiffness -> coupled_bca.wall_stiffness; external_pressure -> coupled_bca.external_pressure; density -> coupled_bca.density; friction_coefficient -> coupled_bca.friction_coefficient; momentum_correction -> cou-
			pled_bca.momentum_correction

Segment/block	Type	Local nodes	Free-parameter fields
coupled_lcca	fixed_3cell_1d_log_area_vessel_block	coupled_lcca_arch; 2: coupled_lcca_out	length -> coupled_lcca.length; reference_area -> coupled_lcca.reference_area; wall_stiffness -> coupled_lcca.wall_stiffness; external_pressure -> coupled_lcca.external_pressure; density -> coupled_lcca.density; friction_coefficient -> coupled_lcca.friction_coefficient; momentum_correction -> coupled_lcca.momentum_correction
coupled_aortic_arch_junction	aortic_arch_total_pressure_junction_block	coupled_dao_arch; 3: coupled_bca_arch; 4: coupled_lcca_arch; 5: coupled_lsa_arch	aao_flow = coupled_aortic_arch_junction.aao_flow; dao_flow = coupled_aortic_arch_junction.dao_flow; bca_flow = coupled_aortic_arch_junction.bca_flow; lcca_flow = coupled_aortic_arch_junction.lcca_flow; lsa_flow = coupled_aortic_arch_junction.lsa_flow; log_area_aao = coupled_aao.log_area_03; log_area_dao = coupled_dao.log_area_01; log_area_bca = coupled_bca.log_area_01; log_area_lcca = coupled_lcca.log_area_01; density -> coupled_aortic_arch_junction.density
coupled_dao_loss	rc_block	1: dao; 2: coupled_dao_out	capacitance -> zero_capacitance; coupled_dao_loss.resistance
coupled_bca_loss	rc_block	1: bca; 2: coupled_bca_out	capacitance -> zero_capacitance; coupled_bca_loss.resistance
coupled_lcca_loss	rc_block	1: lcca; 2: coupled_lcca_out	capacitance -> zero_capacitance; coupled_lcca_loss.resistance

Segment/block	Type	Local nodes	Free-parameter fields
coupled_ivc	fixed_3cell_1d_log_area_vessel; 0: block	coupled_ivc_tpc	length -> coupled_ivc.length; reference_area -> coupled_ivc.reference_area; wall_stiffness -> coupled_ivc.wall_stiffness; external_pressure -> coupled_ivc.external_pressure; density -> coupled_ivc.density; friction_coefficient -> coupled_ivc.friction_coefficient; momentum_correction -> cou-
coupled_rpa	fixed_3cell_1d_log_area_vessel; 1: block	coupled_rpa_tpc; 2: rpa	length -> coupled_rpa.length; reference_area -> coupled_rpa.reference_area; wall_stiffness -> coupled_rpa.wall_stiffness; external_pressure -> coupled_rpa.external_pressure; density -> coupled_rpa.density; friction_coefficient -> coupled_rpa.friction_coefficient; momentum_correction -> cou-
coupled_svc	fixed_3cell_1d_log_area_vessel; 2: block	coupled_svc_tpc	length -> coupled_svc.length; reference_area -> coupled_svc.reference_area; wall_stiffness -> coupled_svc.wall_stiffness; external_pressure -> coupled_svc.external_pressure; density -> coupled_svc.density; friction_coefficient -> coupled_svc.friction_coefficient; momentum_correction -> cou-
			pled_svc.momentum_correction

Segment/block	Type	Local nodes	Free-parameter fields
coupled_lpa	fixed_6cell_tapered_1d_log:area_vessel_block2: lpa		external_pressure -> coupled_lpa.external_pressure; density -> coupled_lpa.density; momentum_correction -> coupled_lpa.momentum_correction; cell_length_01 -> coupled_lpa.cell_length_01; reference_area_01 -> coupled_lpa.reference_area_01; wall_stiffness_01 -> coupled_lpa.wall_stiffness_01; friction_coefficient_01 -> coupled_lpa.friction_coefficient_01; cell_length_02 -> coupled_lpa.cell_length_02; reference_area_02 -> coupled_lpa.reference_area_02; wall_stiffness_02 -> coupled_lpa.wall_stiffness_02; friction_coefficient_02 -> coupled_lpa.friction_coefficient_02; cell_length_03 -> coupled_lpa.cell_length_03; reference_area_03 -> coupled_lpa.reference_area_03; wall_stiffness_03 -> coupled_lpa.wall_stiffness_03; friction_coefficient_03 -> coupled_lpa.friction_coefficient_03; cell_length_04 -> coupled_lpa.cell_length_04; reference_area_04 -> coupled_lpa.reference_area_04; wall_stiffness_04 -> coupled_lpa.wall_stiffness_04; friction_coefficient_04 -> coupled_lpa.friction_coefficient_04; cell_length_05 -> coupled_lpa.cell_length_05; reference_area_05 -> coupled_lpa.reference_area_05; wall_stiffness_05 -> coupled_lpa.wall_stiffness_05; friction_coefficient_05 -> coupled_lpa.friction_coefficient_05; cell_length_06 -> coupled_lpa.cell_length_06; reference_area_06 -> coupled_lpa.reference_area_06; wall_stiffness_06 -> coupled_lpa.wall_stiffness_06; friction_coefficient_06 -> coupled_lpa.friction_coefficient_06;

Segment/block	Type	Local nodes	Free-parameter fields
coupled_tpc_junction	tpc_characteristic_total_pressure_junction; Block	coupled_ivc_tpc; 3: coupled_rpa_tpc; 4: coupled_lpa_tpc	svc_flow = coupled_tpc_junction.svc_flow; ivc_flow = coupled_tpc_junction.ivc_flow; rpa_flow = coupled_tpc_junction.rpa_flow; lpa_flow = coupled_tpc_junction.lpa_flow; log_area_svc = coupled_svc.log_area_03; log_area_ivc = coupled_ivc.log_area_03; log_area_rpa = coupled_rpa.log_area_01; log_area_lpa = coupled_lpa.log_area_01; reference_area_svc -> coupled_svc.reference_area; reference_area_ivc -> coupled_ivc.reference_area; reference_area_rpa -> coupled_rpa.reference_area; reference_area_lpa -> coupled_lpa.reference_area_01; wall_stiffness_svc -> coupled_svc.wall_stiffness; wall_stiffness_ivc -> coupled_ivc.wall_stiffness; wall_stiffness_rpa -> coupled_rpa.wall_stiffness; wall_stiffness_lpa -> coupled_lpa.wall_stiffness_01; external_pressure_svc -> coupled_svc.external_pressure; external_pressure_ivc -> coupled_ivc.external_pressure; external_pressure_rpa -> coupled_rpa.external_pressure; external_pressure_lpa -> coupled_lpa.external_pressure; density -> coupled_tpc_junction.density; wall_pressure_weight -> coupled_tpc_junction.wall_pressure_weight; characteristic_scale -> coupled_tpc_junction.characteristic_scale; loss_coefficient -> coupled_tpc_junction.loss_coefficient

1.7 Free Parameters

Unless a parameter states otherwise in the config, units follow the repository SI convention: pressure in Pa, flow in m^3s^{-1} , resistance in Pa s m^{-3} , capacitance in m^3Pa^{-1} , inertance in $\text{Pa s}^2\text{m}^{-3}$, volume in m^3 , length in m, and time in s.

The entries below are the complete `parameters` dictionary from the authoritative baseline config. Derived-expression entries are shown as compact JSON.

- `aao_compliance.capacitance = 4.0003400289e-09`
- `active_atrium.activation_end = 0.98`
- `active_atrium.activation_peak = 0.9`
- `active_atrium.activation_start = 0.78`
- `active_atrium.elastance_max = 66661000`
- `active_atrium.elastance_min = 16665250`
- `active_atrium.unstressed_volume = 4e-05`
- `active_law.activation.max = 35`
- `active_law.activation.min = -20`
- `arch_lsa.resistance = 7999320`
- `bca_compliance.capacitance = 3.00025502168e-09`
- `capacitance_valve.capacitance = 5e-12`
- `cavity.dynamics.damping_coef = 70`
- `cavity.dynamics.hyperelastic_cst = [444.0,2.9,69.0,6.5]`
- `cavity.dynamics.vol_mass = 1000`
- `cavity.rheology.active_law.activation = {"alpha": "diastole_scaling_factor", "phases": [0,0,1,1,1,0,0], "ref`
- `cavity.rheology.active_law.crossbridge_stiffness = 273000`
- `cavity.rheology.active_law.destruction_rate = 12`
- `cavity.rheology.active_law.starling_abscissas = [-0.1668,-0.0073,0.0534,0.0969,0.1326,0.2016,0.4663,0.91`
- `cavity.rheology.active_law.starling_ordinates = [0.0,0.5614,0.7748,0.8933,0.9618,1.0,1.0,0.1075,0.0]`
- `cavity.rheology.damping_parallel = 70`
- `cavity.rheology.series_stiffness = 100000000`
- `cavity.velocity_law.scheme_ts_hht = 0.4`
- `conduit.scheme_ts_flux = 0.25`
- `coupled_aao.density = 1060`
- `coupled_aao.external_pressure = 0`
- `coupled_aao.friction_coefficient = 0.000111468793255`
- `coupled_aao.length = 0.029`
- `coupled_aao.momentum_correction = 1.1`
- `coupled_aao.reference_area = 0.00033978288044`
- `coupled_aao.wall_stiffness = 3291870.01578`
- `coupled_aortic_arch_junction.density = 1060`
- `coupled_bca.density = 1060`
- `coupled_bca.external_pressure = 0`
- `coupled_bca.friction_coefficient = 8.56474611945e-05`
- `coupled_bca.length = 0.025`
- `coupled_bca.momentum_correction = 1.1`
- `coupled_bca.reference_area = 7.51036993749e-05`
- `coupled_bca.wall_stiffness = 7001849.31528`
- `coupled_bca_loss.resistance = 4930498.77469`
- `coupled_dao.density = 1060`
- `coupled_dao.external_pressure = 0`
- `coupled_dao.friction_coefficient = 0.000101149281846`
- `coupled_dao.length = 0.09`
- `coupled_dao.momentum_correction = 1.1`
- `coupled_dao.reference_area = 0.000137641028135`
- `coupled_dao.wall_stiffness = 5172130.46156`
- `coupled_dao_loss.resistance = 18689019.112`
- `coupled_ivc.density = 1060`
- `coupled_ivc.external_pressure = 0`
- `coupled_ivc.friction_coefficient = 9.68212873625e-05`

- coupled_ivc.length = 0.055
- coupled_ivc.momentum_correction = 1.1
- coupled_ivc.reference_area = 0.000567450173055
- coupled_ivc.wall_stiffness = 11243580.0693
- coupled_lcca.density = 1060
- coupled_lcca.external_pressure = 0
- coupled_lcca.friction_coefficient = 9.08799734509e-05
- coupled_lcca.length = 0.03
- coupled_lcca.momentum_correction = 1.1
- coupled_lcca.reference_area = 2.64090132442e-05
- coupled_lcca.wall_stiffness = 11807755.1202
- coupled_lcca_loss.resistance = 3855594.64542
- coupled_lpa.cell_length_01 = 0.00733333333333
- coupled_lpa.cell_length_02 = 0.00733333333333
- coupled_lpa.cell_length_03 = 0.00733333333333
- coupled_lpa.cell_length_04 = 0.00466666666667
- coupled_lpa.cell_length_05 = 0.00466666666667
- coupled_lpa.cell_length_06 = 0.00466666666667
- coupled_lpa.density = 1060
- coupled_lpa.external_pressure = 0
- coupled_lpa.friction_coefficient_01 = 8.59229888311e-05
- coupled_lpa.friction_coefficient_02 = 8.59229888311e-05
- coupled_lpa.friction_coefficient_03 = 8.59229888311e-05
- coupled_lpa.friction_coefficient_04 = 8.3658541454e-05
- coupled_lpa.friction_coefficient_05 = 8.3658541454e-05
- coupled_lpa.friction_coefficient_06 = 8.3658541454e-05
- coupled_lpa.length = 0.036
- coupled_lpa.momentum_correction = 1.1
- coupled_lpa.reference_area = 4.84443000649e-05
- coupled_lpa.reference_area_01 = 7.4661912908e-05
- coupled_lpa.reference_area_02 = 5.34561624962e-05
- coupled_lpa.reference_area_03 = 3.57847038198e-05
- coupled_lpa.reference_area_04 = 3.15031929985e-05
- coupled_lpa.reference_area_05 = 3.84845100065e-05
- coupled_lpa.reference_area_06 = 4.61639587153e-05
- coupled_lpa.wall_stiffness_01 = 30996947.4487
- coupled_lpa.wall_stiffness_02 = 36632756.0758
- coupled_lpa.wall_stiffness_03 = 44773368.537
- coupled_lpa.wall_stiffness_04 = 47718984.8882
- coupled_lpa.wall_stiffness_05 = 43174319.6607
- coupled_lpa.wall_stiffness_06 = 39420030.9946
- coupled_rpa.density = 1060
- coupled_rpa.external_pressure = 0
- coupled_rpa.friction_coefficient = 8.29854663212e-05
- coupled_rpa.length = 0.012
- coupled_rpa.momentum_correction = 1.1
- coupled_rpa.reference_area = 0.000132732289614
- coupled_rpa.wall_stiffness = 23247710.5865
- coupled_svc.density = 1060
- coupled_svc.external_pressure = 0
- coupled_svc.friction_coefficient = 0.000104039230841
- coupled_svc.length = 0.015
- coupled_svc.momentum_correction = 1.1
- coupled_svc.reference_area = 0.000272631337469
- coupled_svc.wall_stiffness = 16221111.0067
- coupled_tpcj_junction.characteristic_scale = 0
- coupled_tpcj_junction.density = 1060
- coupled_tpcj_junction.loss_coefficient = 2

- `coupled_tpcj Junction.wall_pressure_weight = 0.75`
- `dao_compliance.capacitance = 9.00076506503e-09`
- `diastole_scaling_factor = 0.8`
- `fenestration.resistance = 1.33322e+14`
- `heart_contractility = 37800`
- `heart_radius = 0.02431`
- `heart_rate = 69.9300699301`
- `heart_thickness = 0.0080795`
- `heartbeat_duration = {"factors": [60.0], "inverses": ["heart_rate"], "type": "product"}`
- `ivc_compliance.capacitance = 2.25019126626e-07`
- `lcca_compliance.capacitance = 1.50012751084e-09`
- `left_lung.capacitance = 3.00025502168e-08`
- `left_lung.resistance_1 = 8319292.8`
- `left_lung.resistance_2 = 12478939.2`
- `lower_ca2.capacitance = 2.50021251806e-08`
- `lower_cv2.capacitance = 9.00076506503e-08`
- `lower_ra4.resistance = 43196328`
- `lower_rc2.resistance = 140388066`
- `lower_rv2.resistance = 32397246`
- `lpa_compliance.capacitance = 1.12509563313e-08`
- `lsa_compliance.capacitance = 1.50012751084e-09`
- `pleural.pressure = 0`
- `right_lung.capacitance = 3.00025502168e-08`
- `right_lung.resistance_1 = 5759510.4`
- `right_lung.resistance_2 = 8639265.6`
- `rpa_compliance.capacitance = 1.12509563313e-08`
- `svc_compliance.capacitance = 1.50012751084e-07`
- `upper_bca_to_ca1.resistance = 49595784`
- `upper_ca1.capacitance = 1.50012751084e-08`
- `upper_cv1.capacitance = 6.00051004335e-08`
- `upper_lcca_to_ca1.resistance = 99458212`
- `upper_lsa_to_ca1.resistance = 99458212`
- `upper_rc1.resistance = 156439020.571`
- `upper_rv1.resistance = 67045294.5306`
- `valve_arterial.conductance = 1.3e-05`
- `valve_arterial.inductance = 30000`
- `valve_arterial.scheme_ts_flux = 0.25`
- `valve_atrium.conductance = 9e-06`
- `valve_atrium.inductance = 1000`
- `valve_atrium.scheme_ts_flux = 0.25`
- `valves.backward_conductance = 5e-12`
- `zero_capacitance = 0`

1.8 Documentation and Regeneration

Model-local documentation artifacts:

- `models/coupled_0d_1d/README.md`
- `models/coupled_0d_1d/docs/coupled_0d_1d_schematic.svg`
- `models/coupled_0d_1d/docs/coupled_0d_1d_schematic.png`
- `models/coupled_0d_1d/docs/physioblocks_feasibility.md`
- `models/coupled_0d_1d/docs/one_d_numerics.md`
- `models/coupled_0d_1d/docs/openloop_1d_submodels.md`
- `models/coupled_0d_1d/docs/implementation_notes.md`
- `models/coupled_0d_1d/docs/coupled_0d_1d_technical_reference.md`
- `models/coupled_0d_1d/docs/coupled_0d_1d_technical_reference.pdf`
- `models/coupled_0d_1d/configs/fontan_coupled_0d_1d_smoke.jsonc`
- `models/coupled_0d_1d/configs/fontan_coupled_0d_1d_baseline.jsonc`
- `models/coupled_0d_1d/configs/fontan_coupled_0d_1d_vasodilation.jsonc`

- `models/coupled_0d_1d/configs/fontan_coupled_0d_1d_fenestration.jsonc`
- `models/coupled_0d_1d/configs/fontan_coupled_0d_1d_lpa_obstruction.jsonc`
- `models/coupled_0d_1d/configs/submodel_aorta_1d_openloop.jsonc`
- `models/coupled_0d_1d/configs/submodel_tcpc_1d_openloop.jsonc`
- `models/coupled_0d_1d/configs/submodel_aorta_tcpc_1d_openloop.jsonc`
- `models/coupled_0d_1d/calibration/one_d_openloop_geometry.json`
- `models/coupled_0d_1d/reference_outputs/openloop_1d_validation.json`
- `models/coupled_0d_1d/reference_outputs/smoke_metrics.json`
- `models/coupled_0d_1d/reference_outputs/closed_loop_smoke_validation.json`

Regenerate the technical reference source and PDF with:

```
python3 scripts/docs/build_model_reference_pdfs.py --model coupled_0d_1d
```

1.9 Current Limitations

- The coupled model is executable but not accepted as periodic, calibrated, or clinically validated.
- The aortic total-pressure junction is an algebraic no-loss coupler; the TCPC junction adds branch wall-pressure blending and signed dynamic minor losses but is still not Nektar’s full characteristic/Riemann boundary treatment or a 3-D TCPC loss model.
- The current smoke run is a startup integration test; it is not a physiological cycle validation.
- Atrium and ventricle cycle-balance gates remain failed on the startup smoke output.

The model parameters and standardized data are for computational development and calibration workflows. Simulation outputs must not be presented as clinically validated without separate validation and documentation.